



Autism spectrum disorder diagnosis using fractal and non-fractal-based functional connectivity analysis and machine learning methods

Chetan Rakshe¹ · Suja Kunneth² · Soumya Sundaram³ · Murugappan Murugappan^{4,5,6} · Jac Fredo Agastinose Ronickom¹

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Abstract

Autism spectrum disorder (ASD) is a neurological condition characterized by impaired functional connectivity (FC) networks in the brain. There are several brain networks associated with ASD that have been studied for ASD diagnosis, but the results are inconsistent. A functional magnetic resonance imaging (fMRI) study was performed to address this gap by comparing brain networks among autistic individuals and individuals with typical development (TD) using data from the ABIDE-I and ABIDE-II databases. Blood oxygen level-dependent (BOLD) time series were extracted from 236 regions of interest (ROI) in fMRI data using three atlases: Gordon's, Harvard Oxford, and Diedrichsen. Consequently, 27,730 nonlinear features are extracted from FC matrices, including fractals, non-fractals, and Pearson correlation coefficients (PCC). A parametric and nonparametric classifier was used to analyze the top 0.1%, 0.3%, 0.5%, 0.7%, 1%, 2%, and 3% of features based on the XGBoost feature ranking algorithm. In the study, we found that non-fractal brain FC measures can accurately identify ASD and TD more effectively than fractal and PCC measures. Classifiers performed well, with FC features at the top 0.3%. The classification model at OHSU was more accurate than the model at other sites. There was 100% accuracy at a single site and 96.17% accuracy at all sites using a multilayer perceptron classifier with non-fractal features. The classifier model shows that Cingulo-Parietal Task Control (13.6%), Retro-Splenial Temporal (RST) (13.3%), and Salience (11.6%) are significant contributors. The optimal performance was observed in features derived from networks such as RST and default (4 connections), auditory and Fronto-Parietal Task Control (3 connections), Cingulo-Opercular Task Control (COTC), and ventral attention (3 connections) within COTC (3 connections), visual and COTC (3 connections), and cerebellum and COTC (3 connections). Based on the results, non-fractal-based FC has value in distinguishing ASD from TD using resting-state fMRI.

Keywords Autism spectrum disorder · Resting-state fMRI · Functional connectivity · Fractal · Non-fractal · Machine learning

1 Introduction

The autism spectrum disorder (ASD) is a neurodevelopmental disorder characterized by restricted or repetitive behaviors, as well as atypical social interactions and communication [1]. Across the globe, ASD prevalence varies from region to region, with a median rate of 100 cases per 10,000 people [2]. ASD prevalence varies geographically, with 0.4%, 1%, 0.5%, 1%, and 1.7%, respectively, in Asia, America, Europe, Africa, and Australia [3]. Various studies have suggested that ASD results from the interaction between genes and the environment [4].

Diagnostic criteria for autism include clinical scores and behavioral symptoms; however, behavioral assessments require a highly experienced and subjective multidisciplinary team, which can take weeks or months. Furthermore, it could result in poor treatment due to a lack of understanding of neuropathology [5]. It is important to differentiate typically developing (TD) individuals from those with ASD so that early diagnosis and intervention can improve ASD individuals' outcomes. In addition, researchers may be able to identify neurological distinctions between individuals with TD and individuals with ASD, potentially leading to more effective interventions. It is critical to establish an early and accurate diagnosis

model based on functional or structural brain relationships to distinguish ASD from TD.

As a result of the complexity and heterogeneity of ASD, diagnosis can be challenging due to the impairment of higher-level brain functions and whole-brain structures. In order to address this challenge, several neuroimaging studies have used non-invasive brain imaging techniques to diagnose ASD [6], such as structural magnetic resonance imaging (sMRI) [7], functional magnetic resonance imaging (fMRI) [8, 9], electroencephalography (EEG) [10], magnetoencephalography (MEG) [11], and diffusion tensor imaging (DTI) [12, 13]. In light of these techniques, resting-state fMRI has increasingly become regarded as a more valuable tool for diagnosing ASD than other neuroimaging methods, as it measures blood oxygen level-dependent (BOLD) signals to explore neural mechanisms associated with the disorder [14]. Additionally, fMRI allows for the analysis of neural activity over time with a high level of spatial and temporal resolution. In contrast with sMRI, fMRI focuses more on the brain's functional activities and serves as a better way to assess various brain regions while engaged in certain activities [15]. Consequently, it can be used for identifying potential biomarkers for ASD and understanding the neural mechanisms that underlie it.

Functional connectivity (FC) patterns and microscopic synaptic connectivity have been identified as potential biomarkers for ASD in fMRI studies. FC patterns are derived from resting-state fMRI data by measuring the time-related correlation of BOLD signals among different brain regions. The patterns reveal potential distinctions between individuals with ASD based on the interactions between other brain regions [16]. Many methods have been proposed for constructing brain functional networks through FC modeling, including Pearson correlation coefficient (PCC) [17], Pearson partial correlation (PPC) [18, 19], Spearman's rank correlation coefficient (SRCC) [20], mutual information (MI) [21], and covariance [22]. Likely, conventional correlation methods do not adequately reflect spontaneous neuronal activity dynamics since non-neuronal physiological processes can significantly influence resting-state signals [23]. A fractal pattern of self-similarity and power-law scaling has been observed in both time and frequency domains of fMRI signals during resting states [24–27].

In light of the fact that non-fractal connectivity has an impact on fractal behavior, it may be a new method of measuring FC computed from fMRI. The correlation structure of spontaneous neuronal activity may be more accurately determined by removing the fractal behavior from the BOLD signals than by conventional correlation methods. Due to the nonlinear nature of neurophysiological systems, non-fractal connectivity is not necessarily equivalent to spontaneous neuronal activity correlation. Recent

studies have applied resting-state fMRI fractals and non-fractals to Alzheimer's disease [17] and ASD [28]. A scientific study reported that individuals with ASD had abnormal temporal structures, including an elevated power-law exponent in their salience network [29]. Therefore, we were highly motivated to propose an FC metric based on fractals and non-fractals to distinguish between ASD and TD using resting-state fMRI data and compare it with a conventional FC metric.

The FC matrix derived from different brain regions produces high-dimensional feature vectors. According to a study [30], researchers assessed brain FC in ASD using the Craddock atlas in ABIDE and the autoencoder method for feature reductions. By using deep broad learning, it is possible to reduce low-order and high-level FC features extracted using PCC and hypergraphs, which reduces the computational complexity of machine learning algorithms. In addition, in [31], the investigators used partial and full correlations to extract features from the ABIDE dataset. The researchers used machine learning algorithms, including random forest (RF), conditional RF, oblique RF, support vector machine (SVM), and convolutional neural network (CNN) to classify ASD.

Neuroimaging data have been used in research to improve the diagnosis of ASD by using machine learning [32] and deep learning techniques [33], such as CNN [34] and GNN (graph neural network) [35]. Generally, machine learning models are simpler than deep learning models, making them easier to understand and interpret. In this way, it is possible to interpret the top networks of the brain that contribute to a machine learning model. Additionally, machine learning models are more robust to noise, making them ideal for biomedical applications. Meanwhile, deep learning models are susceptible to noisy data, which can make them unsuitable for real-world applications where imperfect data are common. In machine learning, there are two types of algorithms: parametric and nonparametric. The parameters in a parametric machine learning model are fixed regardless of the number of instances in the training set. Typically, these models require relatively small amounts of data and make assumptions about the data distribution. A nonparametric machine learning model assumes no specific form for the relationship between inputs and outputs, and it can adapt to any functional form in training data. Compared to parametric models, non-parametric models are more flexible, can handle large datasets, and can handle complex relationships. Based on resting-state fMRI data of ASD and TD, researchers in [36] compared linear discriminant analysis (LDA) with non-parametric classifiers like k-nearest neighbor (KNN), RF, and SVM and found RF and SVM were more accurate in ASD diagnosis.

In this present study, the following major contributions are made:

- FC methods such as fractals, non-fractals, and PCC have been explored in the classification of ASD with TD.
- Identified the discriminative brain networks associated with autism spectrum disorders.
- Evaluated the consistency and connections of these brain networks, as well as their underlying neural patterns.

In this study, fMRI was used to investigate brain connections between autistic individuals and individuals with TD. The features were extracted using short covariance techniques based on non-fractal connectivity, asymptotic wavelet correlation, and PCC. By using XGBoost feature ranking, the features were ranked and classified by a variety of machine learning classifiers, including bagging classifiers (BC), KNN, logistic regression (LR), multi-layer perceptrons (MLP), Naïve Bayes (NB), RF, stochastic gradient descent (SGD), and SVM. Compared to state-of-the-art approaches, we achieved a fivefold classification accuracy of 100% using the non-fractal connectivity method and MLP model for OHSU. Furthermore, Cingulo-Parietal Task Control (CPTC) and Retro-Splenial Temporal (RST) networks were found to be more significant in classifying autism spectrum disorders. Additionally, a link between the RST and default mode network proved useful for classification. Machine learning techniques and a distinct workflow pipeline are used in our study to assess the diagnostic categorization of ASD using fractal and non-fractal connectivity methods.

2 Materials and methods

Figure 1 illustrates the pipeline followed in this study. It involves (1) segregating fMRI data of ASD and TD from ABIDE dataset, (2) preprocessing, (3) parcellation and extraction of BOLD time series, (4) computing functional connectivity matrix, (5) feature reduction, and (6) building classification models.

2.1 Dataset

In this study, we used fMRI data from the Autism Brain Imaging Data Exchange I (ABIDE I) [37] and Autism Brain Imaging Data Exchange II (ABIDE II) [38] databases. The dataset includes sMRI, DTI, and resting-state fMRI collected around the world, and all have been approved by local institutional review boards. The ABIDE website provided demographic information and diagnostic status for subjects. The study included data from seven sites that considered various inclusion criteria for resting-state fMRI. Resting-state fMRI data recorded from eyes-

open subjects were included in the analysis because FC is known to be affected by eye state [39]. The age range of our participants was 7–18 years old. In resting-state fMRI, at least 80% of original volumes were retained after filtering, and root mean square deviation (motion) was less than 0.2 to avoid head movement effects on BOLD fluctuations [40]. We recorded participant gender and age based on the demographic information provided in the dataset description. There are 274 males and 37 females among the ASD participants. Conversely, there are 293 males and 107 females with TD. Site-specific demographic information of subjects considering the study is summarized in Table 1.

2.2 Preprocessing

Preprocessing of resting-state fMRI data was performed using the functional neuroimaging analysis software packages AFNI [41] and FSL 5.0 [42]. A series of eight distinct steps were used to enhance data quality in the current study's preprocessing pipeline for fMRI data analysis. The steps are as follows:

- *Trimming*: In this process, the initial fMRI data volumes are removed to achieve T1 equilibrium. Accordingly, the NYU data were corrected by trimming 5 and 3 volumes in ABIDE-I and ABIDE-II datasets, respectively, ensuring T1 signal stability [39].
- *Alignment*: We meticulously aligned functional images with anatomical space using Sinc interpolation and FLIRT. The alignment of functional and anatomical images, achieved through six degrees of freedom, is crucial for correcting discrepancies [31].
- *Normalization*: FNIRT from the FSL suite was used to normalize the aligned images to the MNI152 3 mm template to standardize the images from various sources into a consistent space. The purpose of this step is to address variances in spatial resolution and intensity caused by different MRI scanners.
- *Spatial smoothing*: A key step in enhancing signal-to-noise ratio (SNR) is spatial smoothing. This process aimed for a global full-width-at-half-maximum of 6 mm. Smoothing is carefully balanced to preserve spatial resolution while reducing image noise.
- *Temporal filtering*: A second-order band-pass filter with a passband of 0.008–0.08 Hz effectively removes low-frequency drifts and high-frequency noise from fMRI data [31]. Resting-state fMRI data must be preprocessed using this method to improve clarity.
- *Subject-level regression*: The denoising technique involved subject-level regression against eight nuisance variables and their first derivatives, encompassing six rigid body motion parameters, ventricular cerebrospinal

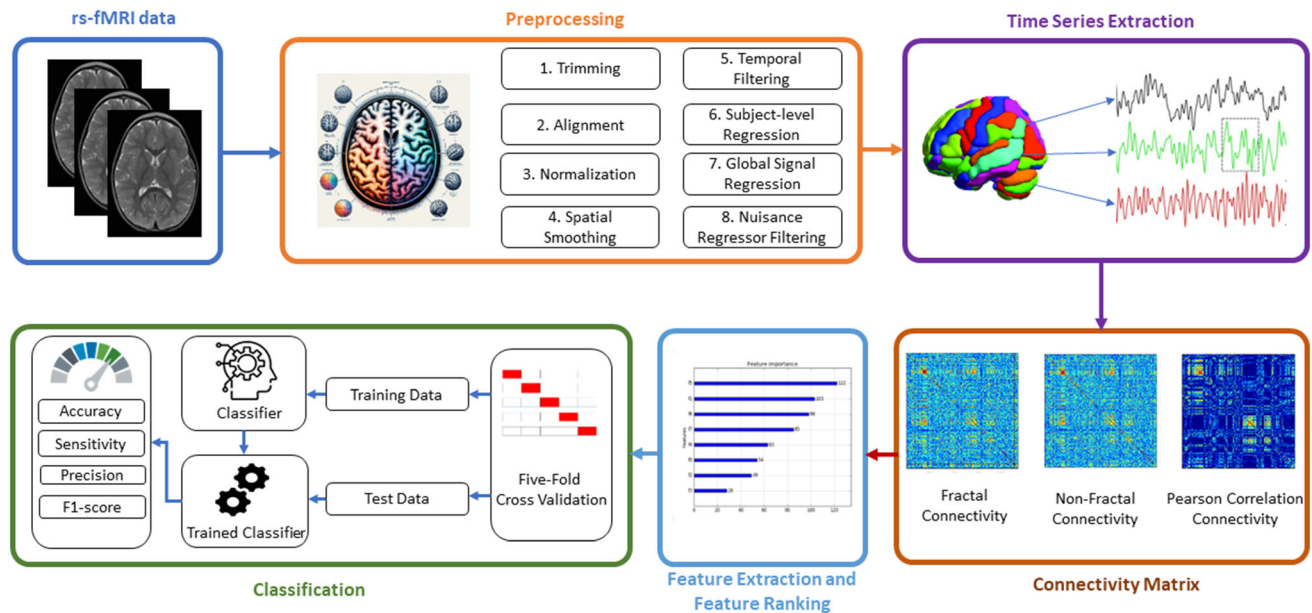


Fig. 1 Process pipeline of the study

fluid, and white matter signals. FSL's FAST method was used to extract the signals, ensuring noise-free resting-state fMRI signals.

- **Global signal regression:** Integrated into the preprocessing pipeline, this step compensates for variations caused by using the data acquired at different sites. Discrepancies in acquisition parameters can be mitigated by using this method, resulting in improved signal-to-noise ratios [43].
- **Nuisance regressor filtering:** A uniform second-order Butterworth band-pass filter with a pass band of 0.008–0.08 Hz was applied to all 17 nuisance regressors to ensure consistency across the dataset [44]. It facilitates more reliable comparisons between subjects and studies because of this uniformity in preprocessing.

2.3 Parcellation and extraction of BOLD time series

In 95% of individuals, BOLD signals were detected in voxels in which the whole-brain mask was created. There were 333 cortical regions of interest (ROIs) used in Gordon's [45], 14 subcortical ROIs from Harvard Oxford [46], 26 cerebellar ROIs from Diedrichsen atlases [47], and a few smaller cerebellar ROIs combined. The choice of atlases in this study was influenced by our previous experience classifying and examining autistic brains [48]. In the next step, the number of voxels in each ROI of the whole-brain mask was determined. In the analysis, ROIs with at least 95% voxels inside the whole-brain mask were considered [31]. A total of 236 ROIs were obtained, of which

213, 14, and 9 were cortical, subcortical, and cerebellar, respectively. The average BOLD signal time series is extracted for each ROI. The signal is averaged across all voxels within a given ROI at every time point, resulting in a representative time series representing the region's overall activity. As a result, the dimensionality of the data has been reduced while maintaining the most relevant information for each region.

2.4 Functional connectivity matrix

We used PCC to compute the FC to distinguish between ASD and TD. Furthermore, wavelet fractal analysis was used to extract fractal and non-fractal FC, and its results were compared with PCC. A connectivity matrix of 236×236 was created for each participant. The 2D correlation feature matrix of size 236×236 resulting from the connectivity methods was flattened to create a total of 27,330 ($236 \times 235/2$) diagnostic features from the lower or upper triangular of the connectivity matrix. Each method produces a $N_s \times N_f$ feature matrix (number of samples $\times$ number of connections). The following subsections describe PCC, fractal, and non-fractal connectivity.

2.4.1 Pearson correlation coefficient

A PCC is calculated by multiplying a covariance between two variables by their standard deviations. The result is always between -1 and 1 and is essentially a normalized measurement of covariance. Let $x_1(t)$ and $x_2(t)$ be stochastic processes where $E[x_1(t)] = \mu_1$ and $E[x_2(t)] = \mu_2$,

then the correlation of $x_1(t)$ and $x_2(t)$ is defined by Eq. (1) [49]

$$\rho_{1,2} = \frac{\text{cov}[x_1(t), x_2(t)]}{(\text{var}[x_1(t)] \cdot \text{var}[x_2(t)])^{\frac{1}{2}}} \quad (1)$$

where

The covariance of x_1 and x_2 is $\text{cov}\{x_1, x_2\} = E\{(x_1 - \mu_1)(x_2 - \mu_2)\}$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x_1 - \mu_1)(x_2 - \mu_2) f_{1,2}(x_1, x_2) dx_1 dx_2 \quad (2)$$

2.4.2 Fractal connectivity

In signal analysis, fractal behavior is the long-range dependence and self-similarity of signals, which can be quantified using the Hurst exponent. Asymptotic wavelet correlation of BOLD signals across brain regions is called fractal connectivity. By comparing two signals over time, asymptotic wavelet correlation measures their similarity. There is a high correlation between two signals only when their values are similar at the same time and on the same time scale. In each time scale, a wavelet transform is applied to the signals to analyze the correlation between their wavelet coefficients. This analysis uses a 6-level discrete wavelet transform, focusing only on approximation coefficients. The wavelet transform was performed using the Daubechies least asymmetric filter of 8 length (*db8*). The coefficients resulting from the two signals were then compared at different time scales to determine if there was a correlation between them. The fractal connectivity ($\hat{Q}_{m,n}$) is calculated by Eq. (3),

$$\hat{Q}_{m,n} = \hat{\xi}_{m,n} \varphi(\hat{d}_m, \hat{d}_n) \quad (3)$$

where

$$\varphi(\hat{d}_m, \hat{d}_n) = \frac{B_1(\hat{d}_m, \hat{d}_n)}{\sqrt{B_1(\hat{d}_m, \hat{d}_m) B_1(\hat{d}_n, \hat{d}_n)}} \cos\left(\frac{\pi}{2}(\hat{d}_m - \hat{d}_n)\right) \quad (4)$$

where $\hat{\xi}_{m,n}$ is non-fractal connectivity as mentioned in the earlier section, B_1 is a parameter that determines the shape of the power spectrum, d_m and d_n are parameters that determine the fractal behavior of the power spectrum [50].

2.4.3 Non-fractal connectivity

BOLD signals are divided into short-term and long-term components during resting states. In a time series, short- and long-term memory describes the temporal dependence between values [51]. Short-memory time series have a high

degree of dependence between values that are close in time, while long-memory time series have a high degree of dependence between values that are far apart in time. Hurst exponents can be calculated from time series to determine short and long memories. Hurst exponents are calculated by raising the range of the cumulative sum to the standard deviation. Power is chosen to maintain an independent relationship between time series scale and ratio. Among the possible Hurst exponent values, values closer to 0 indicate short memory, while values closer to 1 indicate long memory. The short memory covariance between BOLD time-series signals identifies regions of the brain with non-fractal connectivity. Over a short period, it measures the similarity between two signals. The covariance between two signals is large only when their values are similar at the same time (or close in time). The calculation involves considering a window of a fixed width and sliding it along the two signals. For multivariate fractionally integrated noise process $X(t)$ with memory parameter d , and short memory function $u(t)$, non-fractal connectivity ($\xi_{m,n}$) between $x_m(t)$ and $x_n(t)$ is defined as

$$\xi_{m,n} = \frac{\gamma_{m,n}^{(u)}}{\sqrt{\gamma_{m,m}^{(u)} \gamma_{n,n}^{(u)}}} \quad (5)$$

where $\gamma_{m,n}$ denotes the covariance of $u_m(t)$ and $u_n(t)$ given by $\gamma_{m,n} = E[u_m(1)u_n(1)]$ [28]

2.5 Feature selection and classification

There are 27,730 features for each subject in this study based on FC methods. Due to their high computational complexity, it would be impractical to analyze all of these features. The XGBoost feature ranking algorithm was used in our analysis to address this issue [52]. By selecting a specific set of features for comparison purposes, we will be able to analyze the data efficiently and effectively without sacrificing accuracy. Among the ranked features, 0.1, 0.3, 0.5, 0.7, 1, 2, and 3% features were selected for further classification. The performance of the classification model was evaluated using fivefold cross-validation. A fivefold cross-validation was performed to evaluate the performance of the classification model. We divided the samples into five folds and used independent class data to match the samples for motion and age across diagnostic groups. A total of five sets of training data were generated at each site by evenly distributing samples among five folds. A total of four folds of data were used for training, and the remaining fold was used for testing. Supplementary Table 1 shows a visual representation of the folds used for training and testing. The data were manually split across folds to avoid overfitting and bias across folds, ensuring an equal number

of samples from each site and matching ASDs on each fold based on age and RMSDs. The selection of hyper-parameters of these classifiers in their respective libraries is listed in Supplementary Table 2. Supplementary Tables 4 through 8 contain mean, standard deviation, and p -values for ASD and TD groups across folds 1 to 5. Using the Shapiro–Wilk Test, we found that the data are not normally distributed. Therefore, we calculated the corresponding p -values using the Mann–Whitney U Test. ASD and TD have highly similar mean values of age and RMSD, with an insignificant statistical p -value ($p > 0.05$). Furthermore, we have used XGBoost for feature reduction, which further reduces the chances of overfitting the model. Our diagnostic classification of ASD is based on FC features computed from BC, KNN, LR, MLP, NB, RF, SGD, and SVM classifiers. Each classifier is described below:

- **Bagging classifier:** Bagging is an ensemble method, short for bootstrap aggregation. This involves training the same algorithm with different subsets of training data several times. As a result, machine learning algorithms, particularly decision trees, become more stable and accurate.
- **K-nearest neighbors:** This widely recognized supervised machine learning method is primarily used for classification tasks. This algorithm uses the parameter 'k' to determine how many nearest neighbors should be considered. By identifying the shortest distance between each query point and the closest data points in the training dataset, KNN identifies the closest data points within the training dataset. After determining the 'k' closest data points, the algorithm uses a majority voting system to determine the most frequently occurring class among them. Classification for the query point is determined by the class that dominates this vote. As a result of this methodology, KNN is a straightforward yet effective tool for categorizing data based on similarity measures [53, 54]. This study implements a KNN algorithm using Euclidean distance metrics to find the nearest neighbor. The Euclidean distance metrics $d(x, y)$ between two points x and y is calculated using Eq. (6)

$$d(x, y) = \sum_{i=1}^N \sqrt{x_i^2 - y_i^2} \quad (6)$$

where N is the number of features such that $x = \{x_1, x_2, x_3, \dots, x_N\}$ and $y = \{y_1, y_2, y_3, \dots, y_N\}$.

- **Logistic regression:** It is a supervised machine learning algorithm that models the probability of a discrete outcome given an input variable. It essentially uses a logistic function defined below (Eq. 7) to model a binary output variable.

$$f(z) = \frac{1}{1 + e^{-z}} \quad (7)$$

where $z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$, where x_1 to x_n represent the values of the n attributes and β to β_n represents the weights [55].

- **Multi-layer perceptron:** A feed-forward artificial neural network connects layers of neurons in a directed graph from input to output. In contrast with the input neurons, each neuron applies a nonlinear activation function. In backpropagation, the network adjusts its connection weights based on the error in its output versus the expected result. By using this supervised learning process, MLP can map complex input data to outputs, handling complex nonlinear relationships [56]. MLP's key advantage is its flexibility and ability to be implemented like other classifiers. Due to its versatility, MLP is an effective tool in the field of machine learning, where traditional classifiers are applied in various settings where
- **Naive bayes (NB):** It represents a fundamental probabilistic classifier, which operates by applying Bayes' theorem under the assumption of strong independence among features. The simplicity and effectiveness of this classifier make it particularly useful for large datasets. Given a training set D for n different classes and an attribute vector Y along with its associated labels, the class with the highest posterior probability to which attribute Y belongs is given by, where

$$P(C_i | Y) > P(C_j | Y) \text{ for } 1 \leq j \leq n, j \neq i, \quad (8)$$

$$P(C_i | Y) = \frac{P(Y | C_i)P(C_i)}{P(Y)}, \quad (9)$$

Here, $P(Y)$ is the prior probability of Y , $P(C_i)$ represents the class probabilities, $P(C_j | Y)$ is the posterior probability, and $P(Y | C_i)$ is the posterior probability of Y conditioned on C_i [57].

- **Random forest:** This ensemble learning technique creates a large number of decision trees during the training process. Each decision tree is built using a random subset of training data, and it makes its own class prediction. Each tree in the forest provides a class prediction when a new data point needs to be classified. As a result, the RF algorithm aggregates these predictions and selects the most frequently predicted class. According to the RF model, the majority vote or 'mode' of the classes determined by the individual trees is the final output. With this approach, prediction accuracy and robustness are enhanced because multiple decision trees are used to make decisions. This reduces the risk of errors associated with using a single tree [54].

- *Stochastic gradient descent*: This method is effective and straightforward, especially when the loss function is convex, and it is primarily used to optimize linear classifiers and regressors. In models such as linear SVM and LR, loss functions are commonly used. By iteratively adjusting the model parameters, SGD minimizes the loss function, which measures the difference between the predicted and actual outcomes [58]. In SGD, the model parameters (weights) are updated using the equation (Eq. 10) given below:

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \cdot \nabla_{\theta} J(\theta; x^{(i)}, y^{(i)}) \quad (10)$$

where θ represents the parameters (or weights) of the model, θ_{old} is the current value of the parameters, θ_{new} is the updated value of the parameters, η is the learning rate, a hyperparameter that determines the step size at each iteration, and $\nabla_{\theta} J(\theta; x^{(i)}, y^{(i)})$ is the gradient of the objective function J with respect to the parameters θ . This gradient is computed using a single data point (or a mini-batch), denoted by $x^{(i)}$ and $y^{(i)}$, where $x^{(i)}$ is the feature vector of the i th sample and $y^{(i)}$ is its corresponding label [59].

- *Support vector machines*: It is a widely used supervised machine learning algorithm known for its versatility in both classification and regression tasks. SVM's primary goal is to identify the optimal hyperplane that acts as a decision boundary, effectively separating data points belonging to different classes. In classification problems, SVM attempts to find the hyperplane with the greatest margin or the greatest distance between the hyperplane and the nearest data point from each class. The optimal hyperplane that separates the data can be written as follows if the two classes can be linearly separated (Eq. 11),

$$g(x) = w^T * x + b = 0 \quad (11)$$

where w is a normal vector that is perpendicular to the hyperplane and b is the offset of the hyperplane. In the training procedure, the SVM optimizes w and b to maximize the margin (or distance) between the parallel hyperplanes while still separating the data [60]. This approach enhances the model's ability to generalize well to new data, reducing the risk of overfitting.

We assessed each classifier's effectiveness across various aspects using metrics such as accuracy, precision, recall, and F1-score [61]. The performance of classifiers was compared based on the default parameters of the scikit-learn library based on accuracy. The experiments were carried out on an Intel(R) Core (TM) i7-10,700 processor running at 2.90 GHz and 16 GB of RAM in the Dell Vostro 3888 workstation. A Windows 10 Pro operating system was installed on the workstation. Our FC matrix

calculations were performed using MATLAB (Version R2017b), and classification was conducted using Python 3.9. In the following section, we will (i) evaluate the diagnostic efficacy of the utilized classifiers for autism, (ii) analyze the methodologies adopted for brain connection estimation and their implications, (iii) investigate the significance of specific brain regions and features in the autism diagnostic process, and (iv) examine how brain connectivity analyses can identify key brain regions relevant to autism spectrum disorders diagnosis.

3 Experimental results and discussion

3.1 Brain connection estimators and their effects

FC computed using PCC, fractal, and non-fractal methods for a TD subject is shown in Fig. 2 (a-c). Each method produced a 236 × 236 matrix since we considered 236 regions in our analysis. This matrix shows the correlation between resting-state fMRI signals for different brain regions. PCC, fractal, and non-fractal matrix diagonal values are 1, indicating self-correlation between the regions. Due to the different methods used to calculate the correlation matrix, the patterns between PCC and fractal were very different. There is, however, a small difference between the correlation matrix of the fractal method and the non-fractal method. A memory parameter determines the difference between fractal connectivity and non-fractal connectivity; when the memory parameter difference is small, the two are nearly identical.

Non-fractal connectivity can be explained by removing non-physiological features from fMRI data. FC matrices based on PCC, fractal, and non-fractal methods for an autistic individual are shown in Fig. 3a–c. There are distinct differences between individuals with TD and individuals with autism, as revealed by the correlation matrices. TD exhibits strong connections in some regions, as indicated by the red color, whereas autistic subjects do not show such patterns. The FC matrices of all three methods considered in the study exhibit this feature. Meanwhile, it is possible that we are not able to detect similar patterns across all samples visually.

3.2 Analyzing the effects of classifiers

A feature vector was constructed from the 236 × 236 connectivity matrices of each method, resulting in 27,730 features per subject. We ranked features using XGBoost, built site-specific classification models based on the top percentages of features, and compared their performance. The data in Fig. 4a–c show the fivefold average classification accuracy across the sites considered for different

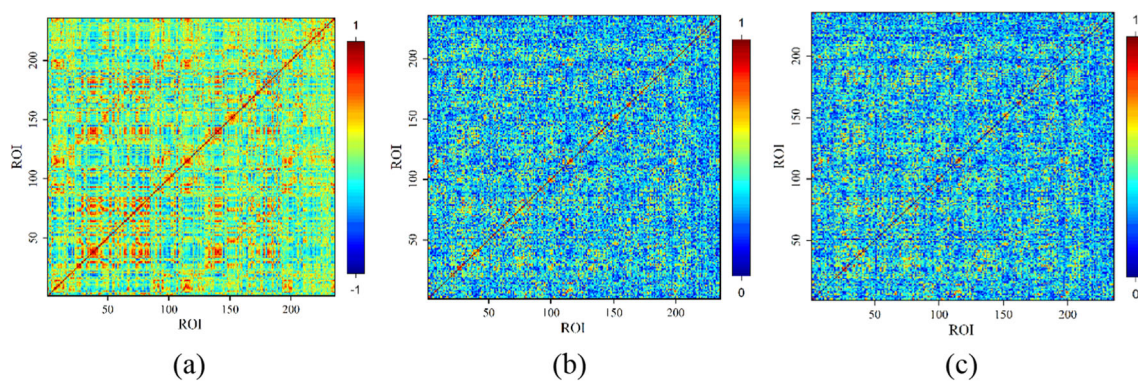


Fig. 2 Brain FC in TD using **a** PCC, **b** fractal, and **c** non-fractal (x and y axis refers to 236 ROIs listed in Supplementary Table 3)

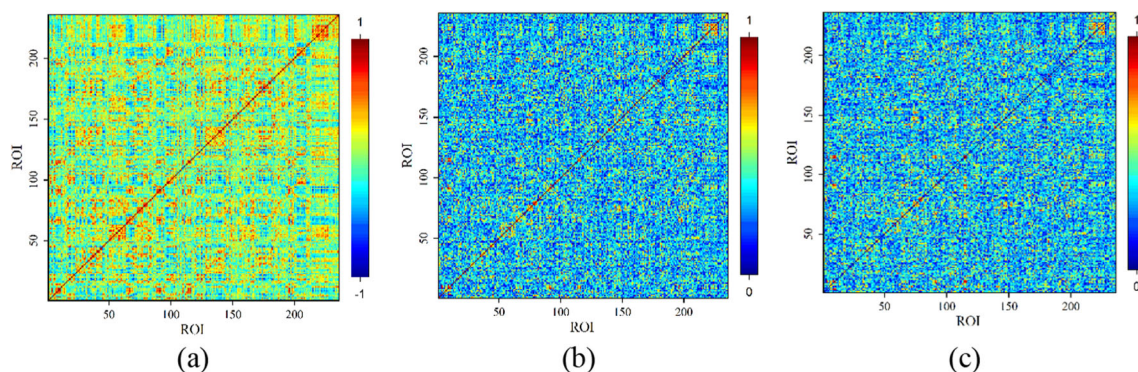


Fig. 3 Brain FC in ASD using **a** PCC, **b** fractal, and **c** non-fractal (x and y axis refers to 236 ROIs listed in Supplementary Table 3)

classifiers based on PCC, fractal, and non-fractal top feature distributions. There are small differences between these methods, but their performance trends are similar. As shown in Fig. 4a, SGD, NB, MLP, and KNN classifiers achieve their maximum accuracy at 0.3% of PCC features. There is an unusual behavior in RF and BC, which peaks at 0.1% and then gradually declines. Comparatively, the LR classifier provides high classification accuracy since only 0.5% of the features contribute to classification accuracy. Figure 4b shows the accuracy trend for fractal correlation features. Most of the classifier's peak accuracy is accounted for by the top 0.3% of features, followed by the top 0.5%. In addition, smaller features (0.1%) and larger features (2% and 3%) are poorly classified by most classifiers, except for the RF and BC. The MLP performed the best on average, with an accuracy of 0.3% for top features. In the same way, Fig. 4c shows non-fractal connectivity following a similar pattern. KNN, LR, and MLP classifiers achieve peak accuracy at 0.3% of top features, while BC, SVM, NB, and SGD classifiers achieve it at 0.5%. In contrast, RF classifiers achieve peak accuracy at 0.1% of top features and then gradually decrease. It was observed that both a small number of features and a large number of features had low accuracy. In general, MLP classifiers achieve higher accuracy at 0.3% of top features.

Based on their accuracy, we compared PCC, fractal, and non-fractal FC techniques, along with seven different classifiers. Figure 4a–c illustrates that all three techniques exhibit similar patterns of classifier performance despite each method having its own peak accuracy. MLP demonstrated the highest accuracy for both fractal and non-fractal connectivity, whereas LR demonstrated the highest accuracy for PCC. A comparison of the performance of LR, SVM, and MLP was possible, but there was a lack of accuracy in RF and BC in all cases.

Studies using SVM and RF have reported higher classification accuracy for ASD diagnosis [8, 16]. Studies have shown that LR [62] and MLP [63] can perform better at classifying ASDs than SVMs. According to our results, OHSU and UCLA data showed better performance than other sites. Several studies using fMRI data to classify ASD by site have reported similar positive results [63, 64]. Additionally, we found that classifiers did not perform well when there were both a low number of features and a high number of features. Most classifiers experienced low accuracy when they used either a small or large number of features, but maximum accuracy was achieved when the number of features was optimal. The majority of discriminative components may not be present in lower numbers of features. The accuracy levels for the top 0.3% and 0.5%

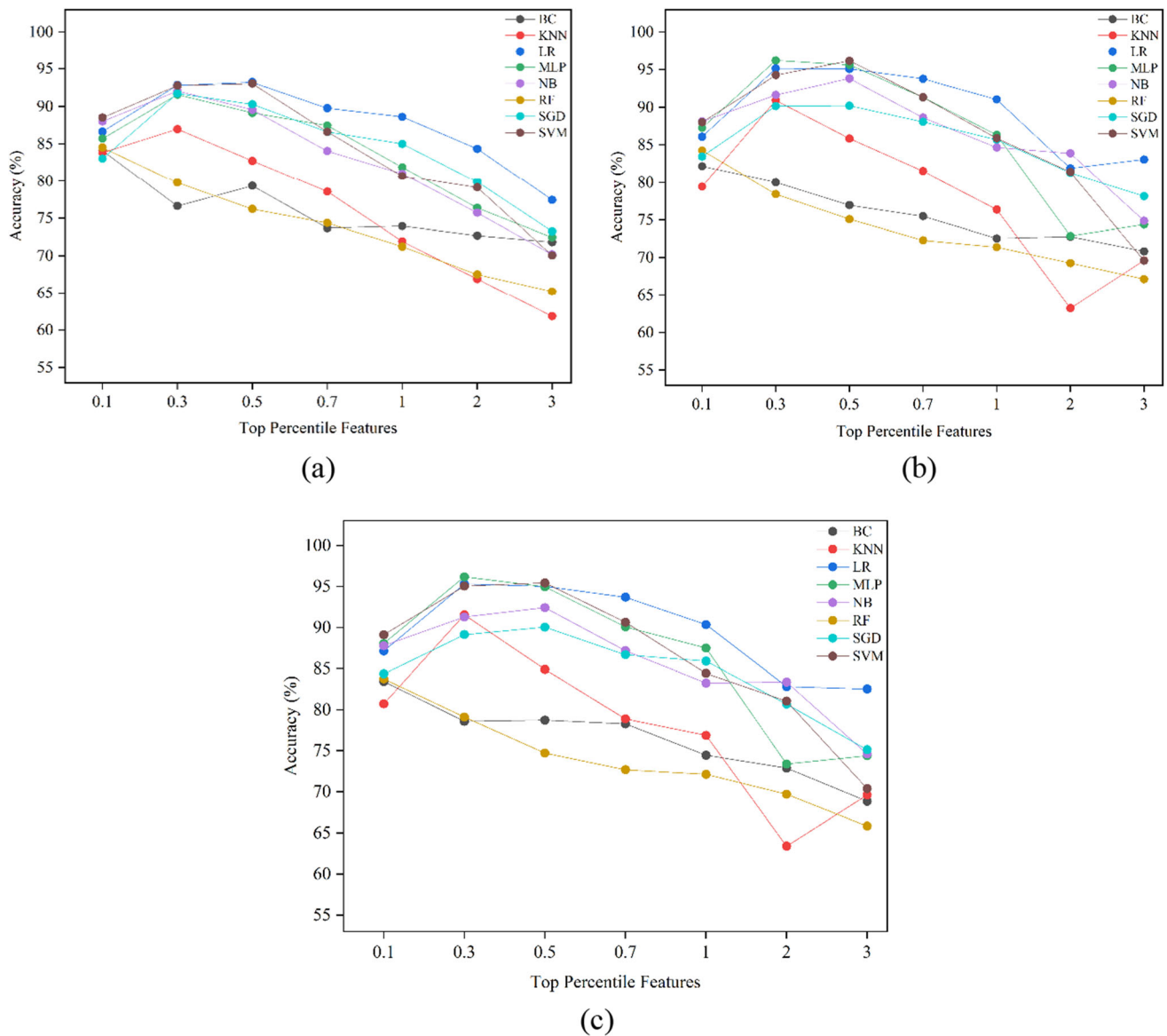


Fig. 4 Performance of classifiers on different dimensions of features **a** PCC **b** fractal and **c** non-fractal

of features (approximately 83 and 138 features) were similar, while accuracy declined with an increasing number of features. A trade-off seems to exist between the number of features and classifier performance. However, most discriminative components have already been accounted for, and more features may include insignificant patterns [65]. A machine learning model with the optimal number of features will achieve the best performance. Several machine learning models have been developed for diagnosing ASD that follow a similar pattern with different features [8, 64, 65]. However, the number of features necessary to build a successful classifier varies significantly between studies. The reasons may be due to the utility of different sample counts, atlases, and feature reduction techniques.

3.3 Importance of different sites in the brain for ASD diagnosis

Figure 5a–c shows the classification accuracy of seven considered sites using PCC, fractals, and non-fractal connectivity methods. According to our results, FC methods and machine learning algorithms performed similarly at OHSU and UCLA. In the fractal method, UCLA had the highest accuracy, while in PCC and non-fractal methods, OHSU had the highest accuracy. There may be a correlation between the number of autistic children and the number of typical development children in these regions. In addition, both ASD and TD had similar amounts of data from male and female groups. Figure 5a shows that FC features computed by the PCC from OHSU's fMRI data

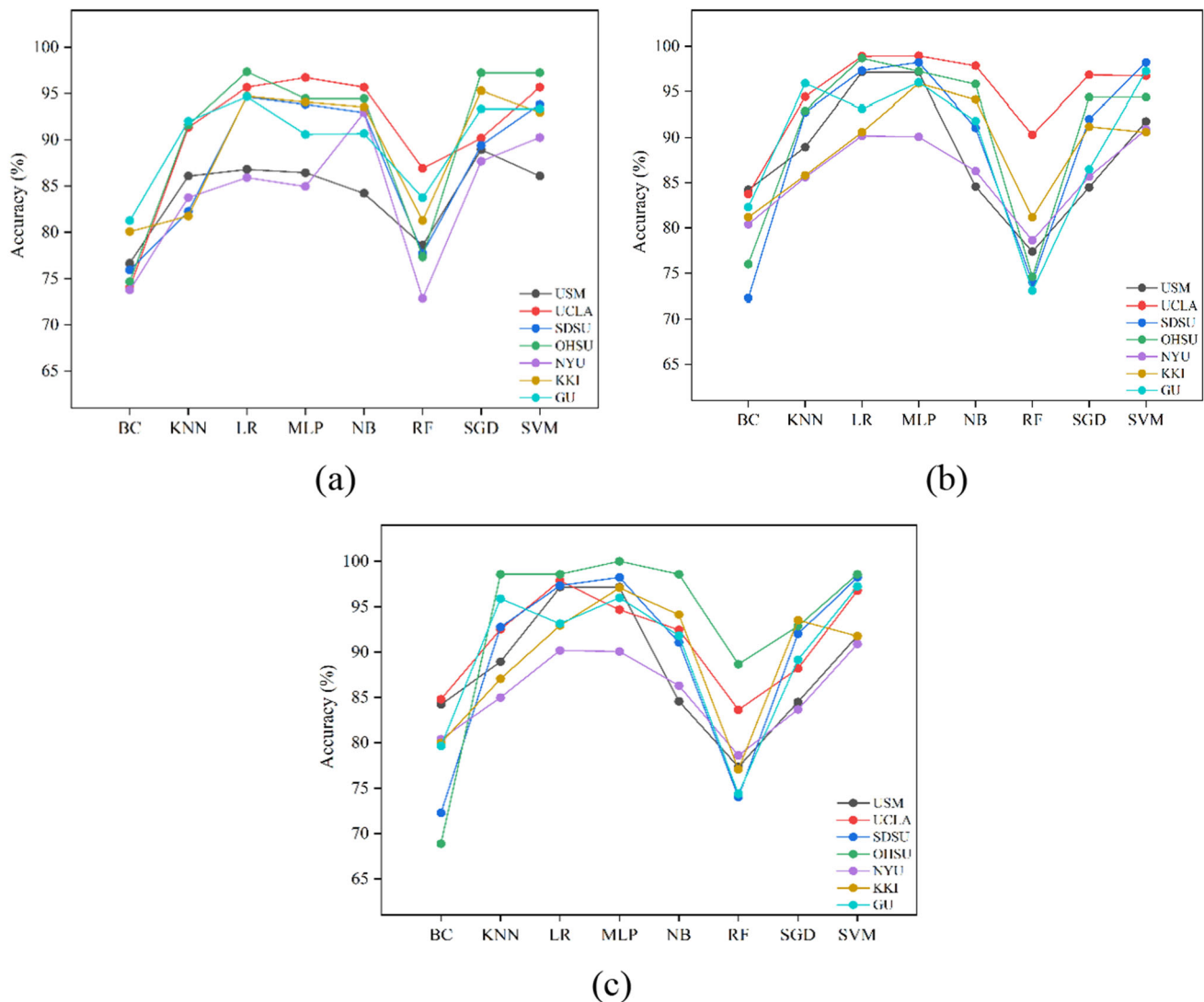


Fig. 5 Performance of classifiers on different sites **a** PCC **b** fractal and **c** non-fractal

were well classified, followed by UCLA's and KKI's. According to the LR, the OHSU site scored a high average fivefold classification accuracy of 97.33%, while SGD and SVM scored 97.23%. The average accuracy of LR was also the highest among all sites. Of all the classifiers, BC and RF had the lowest accuracy. A comparison of classifier performance using fractal correlation features is shown in Fig. 5b. It was found that MLP, LR, and SVM were the most accurate, whereas BC and RF were the least accurate. Additionally, the MLP classifier performed best for the UCLA site, with an accuracy rate of 98.94%. However, the LR classifier performed well for both UCLA (98.88%) and OHSU (98.66%). LR has also been identified as the classifier with the highest average classification accuracy. Figure 5c shows the performance of classifiers for different sites based on non-fractal correlation features. According to OHSU's results, MLP classifiers achieved 100%

accuracy, followed by KNN, LR, NB, and SVM, with an accuracy of 98.57%. The MLP classifier also produced the highest average accuracy across all sites. The accuracy of BC and RF was the lowest among all classifiers.

Based on fMRI data from OHSU and UCLA, literature reports indicate significantly higher performance in the classification of ASDs [63, 64]. Classifiers for the NYU site failed to perform well according to all three methods of feature extraction. A comparison of the proposed model with other state-of-the-art methods is presented in Table 2. In the past, most studies focusing on ASD diagnosis have examined only one site and have not considered brain connectivity patterns. Recently, few researchers have examined brain functional connectivity patterns for multiple site-based ASD diagnoses [30, 62, 63, 66, 67]. Additionally, between 2000 and 2015, fMRI-based resting-state ASD diagnosis through functional connectivity patterns

Table 1 Site-specific demographic information

Sites	TD		ASD	
	Count	Age (Mean $\pm$ SD)	Count	Age (Mean $\pm$ SD)
GU	40 (M 22, F 18)	10.69 $\pm$ 1.61	34 (M 30, F 4)	11.28 $\pm$ 1.3
KKI	125 (M 88, F 37)	11.17 $\pm$ 2.46	45 (M 32, F 13)	12.00 $\pm$ 3.05
OHSU	35 (M 22, F 13)	10.50 $\pm$ 1.70	36 (M 30, F 6)	11.80 $\pm$ 2.30
SDSU	57 (M 42, F 15)	13.22 $\pm$ 2.77	55 (M 47, F 8)	13.57 $\pm$ 2.65
UCLA	47 (M 38, F 9)	12.63 $\pm$ 3.02	45 (M 43, F 2)	12.63 $\pm$ 2.88
NYU	75 (M 60, F 15)	11.31 $\pm$ 2.65	78 (M 74, F 4)	11.80 $\pm$ 3.01
USM	21 (M 21, F 0)	14.19 $\pm$ 3.78	18 (M 18, F 0)	14.88 $\pm$ 2.68

M: male, *F*: female, *SD*: standard deviation, *GU* (Georgetown University), *KKI* (Kennedy Krieger Institute), *OHSU* (Oregon Health and Science University), *SDSU* (San Diego State University), *UCLA* (University of California, Los Angeles), *NYU* (NYU Langone Medical Center), *USM* (University of Utah School of Medicine)

Table 2 Site-specific performance comparisons between the proposed methods and competing methods which used ABIDE dataset

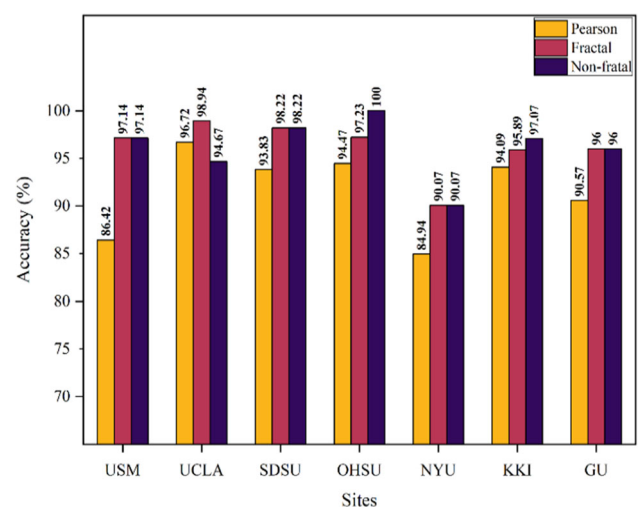
References	Classifiers	FC	Classification rate in %						
			KKI	NYU	OHSU	SDSU	UCLA	USM	GU
[30]	Deep broad learning	Pearson	62.5	71.4	80.8	61.6	79.5	81.6	–
[62]	DNN ^a	PCC	69	68	82	63	73	63	–
[63]	SVM	PCC	–	57.1	54	–	55.1	64.7	–
	SVM-ATM		–	69.1	72.3	–	72.2	69.6	–
	MLP		–	68.5	64	–	71.9	64	–
	MLP-DA ^b		–	70	74.3	–	72.7	70	–
	MLP-SVM-ATM ^c		–	69.7	78	–	70.6	72.3	–
	Auto-ASD-Network		–	70	80	–	72.2	72.4	–
[66]	KNN	PCC	95.8	79.4	94.6	91.6	83.6	88.7	–
[67]	Few shots learning and Simaese network	PCC	66.66	–	57.14	–	–	–	–
Ours	MLP	Non-fractal	97	90	100	98	94	97	97

^aDNN: Deep neural network, ^bDA: Data augmentation, ^cATM: Auto-tuned models

was studied [68]. According to Table 2, the proposed methodology is more effective and comparable to previous works at detecting ASD based on features extracted from seven different sites. This study confirms that nonlinear features are more effective at detecting ASD than other statistical features.

3.4 Importance of features in ASD diagnosis

Figure 6 illustrates the performance of MLP classifiers with optimal features extracted from PCC, fractal, and non-fractal methods. In comparison with the other sites, OHSU was more capable of distinguishing between ASD and TD. According to MLP, the non-fractal method produced 100% accuracy for the OHSU site, and the fractal connectivity method produced 98.88% accuracy for the UCLA site. In some sites (USM, SDSU, GU, NYU), the fractal

**Fig. 6** Performance of MLP classifier on different sites using optimal PCC, fractal, and non-fractal features

connectivity method and non-fractal connectivity method were similar in classification accuracy. There was no significant difference between fractal and non-fractal connectivity features for these sites. Additionally, the non-fractal method performed better than the fractal connectivity method at KKI and OHSU but slightly worse at UCLA. In general, the non-fractal method performed better than both the PCC and the fractal method. The NYU site shows poor performance with MLP classifiers. There may be a difference in data acquisition protocol between this site and others, causing this variation in performance.

In Table 3, the contribution of different brain networks is summarized using FC methods, including PCC, fractal, and non-fractal. In terms of feature contributions, there were differences across the sites (UCLA, OHSU, GU, KKI, SDSU, USM, and NYU). In comparison with the fractal and non-fractal methods, the PCC method doesn't produce the same results. Several sites showed similar performance for fractal or non-fractal methods, which was reflected in machine learning models (Fig. 6). There was a greater degree of consistency across different sites and methods of brain networks, such as the CPTC and RST. The non-fractal method produced the highest accuracy at OHSU, with FC features from CPTC (13.6%), RST (13.33%), and

Salience (11.66%) contributing the most. Based on the fractal connectivity method, UCLA achieved the highest accuracy using SSMM (16.7%), CPTC (12.52%), and default (9.22%). In all three FC methods considered, RST, CPTC, and default networks contributed the most to the GU site's features.

Based on our findings, the non-fractal method outperforms the fractal method, and both methods outperform the PCC method. A similar observation was reported when comparing the effectiveness of these methods in diagnosing ASD [28] and Alzheimer's disease [17]. According to our research, non-fractal methods are superior to fractal methods in distinguishing differences between brain connectivity between individuals with autism and those with TD. Moreover, fractal dimension analysis has been used to diagnose ASD using fMRI and sMRI [50, 69].

3.5 Importance of brain network connections in ASD diagnosis

Table 4 summarizes the number of connections between brain networks based on different methods. There are significant differences between connections to conventional PCC, fractal correlation, and non-fractal correlation across

Table 3 Consistency of the brain networks between the sites by different FC methods (only the top three networks are listed here)

Sites	PCC	Fractal	Non-fractal
USM	SSMM (16.98%)	RST (25.52%)	RST (25.52%)
	RST (9.70%)	CPTC (21.36%)	CPTC (21.36%)
	None (9.43%)	Visual (11.78%)	Visual (11.78%)
UCLA	RST (13.95%)	SSMM (16.7%)	CPTC (13.78%)
	CPTC (8.87%)	CPTC (12.52%)	Default (11.6%)
	Default (7.94%)	Default (9.22%)	Salience (9.18%)
SDSU	Salience (11.20%)	Salience (14.29%)	Salience (14.29%)
	None (10.95%)	None (10.58%)	None (10.58%)
	RST (8.96%)	Default (10.03%)	Default (10.03%)
	SSMM (8.96%)		
OHSU	RST (21.45%)	CPTC (18.26%)	CPTC (13.6%)
	CPTC (17.61%)	Salience (12.17%)	RST (13.33%)
	Visual (9.43%)	RST (11.59%)	Salience (11.66%)
NYU	SSMM (16.98%)	Cerebellum (16.84%)	Cerebellum (16.84%)
	RST (9.70%)	DA (11.22%)	DA (11.22%)
	None (9.43%)	VA (9.20%)	COTC (9.07%)
KKI	CPTC (12.26%)	DA (10.62%)	DA (10.95%)
	Default (10.84%)	Auditory (9.95%)	Auditory (9.9%)
	Salience (9.81%)	Default (9.50%)	RST (9.05%)
GU	RST (19.31%)	RST (20.50%)	RST (20.50%)
	CPTC (13.91%)	CPTC (16.88%)	CPTC (16.88%)
	Default (10.88%)	Default (11.10%)	Default (11.10%)

(Somato-sensory motor hand (*SMhand*), cingulo-opercular task control (*COTC*), somato-sensory motor mouth (*SSMM*), ventral attention (*VA*), dorsal attention (*DA*))

Table 4 Consistency of the brain connections between the sites by different FC methods (only connections between the brain networks higher than 2 are summarized here)

Sites	PCC	Fractal	Non-fractal
USM	Visual—Default (8 connections)	Visual—Default (9 connections)	Visual—Default (9 connections)
	Visual—Visual (7 connections)	Visual—Visual (6 connections)	Visual—Visual (6 connections)
	Visual—RST (6 connections)	Visual—RST (6 connections)	Visual—RST (6 connections)
UCLA	Visual—FPTC (4 connections)	Auditory—Default (3 connections)	Subcortical—COTC (4 connections)
	Cerebellum—DA (4 connections)	VA—SMhand (3 connections)	Auditory—Default (3 connections)
	Visual—Default (3 connections)	DA—DA (3 connections)	Visual—COTC (3 connections)
	Subcortical—COTC (3 connections)	Subcortical—COTC (3 connections)	
SDSU	COTC—Visual (4 connections)	Default—COTC (5 connections)	Default—COTC (5 connections)
	COTC—COTC (4 connections)	COTC—SMhand (4 connections)	COTC—SMhand (4 connections)
	Cerebellum—COTC (4 connections)	DA—COTC (3 connections)	DA—COTC (3 connections)
		Default—FPTC (3 connections)	
OHSU	Visual—Default (5 connections)	RST—Default (3 connections)	RST—Default (4 connections)
	RST—Default (4 connections)	Visual—RST (3 connections)	Auditory—FPTC (3 connections)
	COTC—COTC (4 connections)	COTC—Visual (3 connections)	COTC—VA (3 connections)
	DA—Visual (3 connections)	DA—COTC (3 connections)	COTC—COTC (3 connections)
	Visual—COTC (3 connections)	COTC—COTC (3 connections)	Visual—COTC (3 connections)
NYU		Cerebellum—COTC (3 connections)	Cerebellum—COTC (3 connections)
	FPTC—COTC (4 connections)	COTC—COTC (5 connections)	COTC—COTC (5 connections)
	SMhand—SMhand (3 connections)	Visual—COTC (4 connections)	Visual—COTC (4 connections)
	SMhand—Visual (3 connections)	Cerebellum—Auditory (4 connections)	Cerebellum—Auditory (4 connections)
	VA—SMhand (3 connections)		
KKI	RST—Default (4 connections)	DA—DA (4 connections)	DA—DA (4 connections)
	Visual—Default (3 connections)	DA—SMhand (4 connections)	DA—SMhand (4 connections)
	Visual—FPTC (3 connections)	Auditory—Visual (4 connections)	Auditory—Visual (4 connections)
	RST—Auditory (3 connections)		
GU	COTC—COTC (3 connections)		
	COTC—Default (5 connections)	Visual—Default (4 connections), RST—Default (4 connections), Visual—FPTC (3 connections),	Visual—Default (4 connections)
	COTC—DA (4 connections)	Visual—Visual (3 connections),	RST—Default (4 connections)
	COTC—Visual (3 connections)	SMhand—SMhand (3 connections)	Visual—FPTC (3 connections)
	FPTC—SMhand (3 connections)		Visual—Visual (3 connections)
	Subcortical—DA (3 connections)		SMhand—SMhand (3 connections)

sites. According to our results, fractal FC and non-fractal FC contributed different amounts of brain connections and networks (OHSU, UCLA, and SDSU). However, no difference was found between KKI, NYU, and SDSU. It can be noted that the non-fractal helped to produce high classification accuracy in the site OHSU, and most brain connections occurred between the networks, for example, RST to default (4 connections), auditory to FPTC (3 connections), COTC to VA (3 connections), within COTC (3 connections), visual to COTC (3 connections), and cerebellum to COTC (3 connections). It was not possible to find a consistent pattern among the brain connections among the sites. In the PCC-FC method, connections within

COTC were consistent across three sites (KKI, OHSU, SDSU), and a visual to default network was consistent across all four sites (KKI, OHSU, UCLA, USM). In the case of fractal FC, connections between visual to default, within DA, within COTC, visual to RST, RST to default, within COTC, and visual were similar for two sites (GU, USM), (UCLA, KKI), (OHSU, SDSU), (OHSU, USM), (GU, OHSU), (NYU, OHSU) and (GU, USM). In the non-fractal method, connections between NYU, OHSU, and UCLA's visual and COTC networks were identified as the top connections.

In Tables 3 and 4, the consistency of brain networks and connections between different sites is compared using

different FC methods. In our study, we found that different brain networks contributed to different classification models. Additionally, the top-performing network and percentage of features contributed to site-specific classification models varied by the FC method [70]. There is a noticeable difference between PCC's top brain networks and the other two methods, which may explain its improved performance. Classification accuracy was also affected by small differences, even when some sites were capable of distinguishing between fractal and non-fractal connectivity. In a previous study [31], we examined the effect of FC methods on different cross-validation trials. However, no studies have compared brain networks across sites and FC methods on site-specific classification models. According to our analysis of site data from OHSU, the CPTC network contributed 13.6% of the features to the classification model. As a result of ASD, this network plays a crucial role in a variety of executive functions such as attention, salience, social cognition, and communication. Further evidence of the relevance of these networks in ASD is provided by studies by Agastinose Ronicko et al. [31] and Chen et al. [71], which identified critical alterations in resting-state networks centered in the cingulate for distinguishing people with ASD from people with TD. In addition, our analysis highlighted the RST network's important contribution to our classification model, with 13.33% of features coming from it. Researchers have found that the retrosplenial cortex is crucial for several cognitive processes, such as spatial memory and navigation, as well as imagination and planning for the future [72]. According to Yerys et al., FC within the ventral attention and retrosplenial systems correlates significantly with ASD symptom severity [73]. The retrosplenial cortex is significantly more connected to the insula and dorsolateral prefrontal cortex than the insula and lateral prefrontal cortex, according to [74]. Furthermore, FC from the salience network contributes 11.66% of features to the classification models. The salience network plays a broad role in the management of cognitive control tasks, including monitoring conflicts, processing the importance of stimuli, resolving interferences, maintaining and implementing task strategies, and structuring behavioral responses [75]. Researchers found unusual patterns of FC in ASD individuals, especially within the salience network and between the salience network, the default mode network, and the central executive network [76]. According to Chen et al., individuals with ASD show increased connectivity between their medial prefrontal cortex and their insula, which constitutes part of the salience network [71].

3.6 Brain network contributions and connectivity patterns of mlp model for the OHSU site

According to Fig. 7a, different brain networks contributed to the performance of the MLP classifier at OHSU. According to this study, CPTC, RST, salience, and FPTC all contributed significantly to classification. Some of the features of classification were contributed by the subcortical, none, and dorsal attention networks of the brain. Figure 7b shows a 3D brain map showing the anatomical locations of significant brain connections. A three-dimensional (3D) brain map was generated using the BrainNet Viewer toolbox in MATLAB [77]. The 3D map shows that 83 connections contributed to high classification accuracy. The trans-axial image shows that the left hemisphere of the brain has more connections than the right. The connectomes of top networks used at OHSU to classify using a non-fractal connectivity method and MLP classifier are shown in Fig. 8. The thickness of the cord can be used to predict the rate of occurrence of connections. Thus, the link between RST and default in categorization had a greater significance since it occurred more frequently. Additionally, there is a significant connection between auditory and FPTC, COTC, and ventral attention, within COTC, and visual and cerebellum to COTC.

3.7 Significance of the proposed methodology in ASD Diagnosis

In Table 5, we summarize the findings of studies using resting-state fMRI and sMRI for diagnosing autism spectrum disorders. In most early studies, resting-state fMRI was used, and very few studies combined sMRI and fMRI to diagnose autism spectrum disorders. ASD diagnosis was based on resting-state fMRI data that were utterly inconsistent in earlier studies. Most studies used a single ATLAS, only one site (BASC/AAL), and a limited number of regions of interest (mostly 116 ROI) and different numbers of subjects from the ABIDE I and ABIDE II datasets to diagnose ASD. In addition, brain FC patterns have not been taken into account when analyzing the impact of different brain regions on ASD diagnosis. The resting-state fMRI data have been analyzed using preprocessing, feature extraction, feature selection, and classification approaches by only a few researchers. Among the most widely used classifiers for ASD diagnosis, KNN, SVM, and RF are most commonly used.

In this study, 168 different combinations of process pipelines were tested using seven site data sets, three FC methods, and eight classification models. Based on our analysis, the processing pipeline involving OHSU site data,

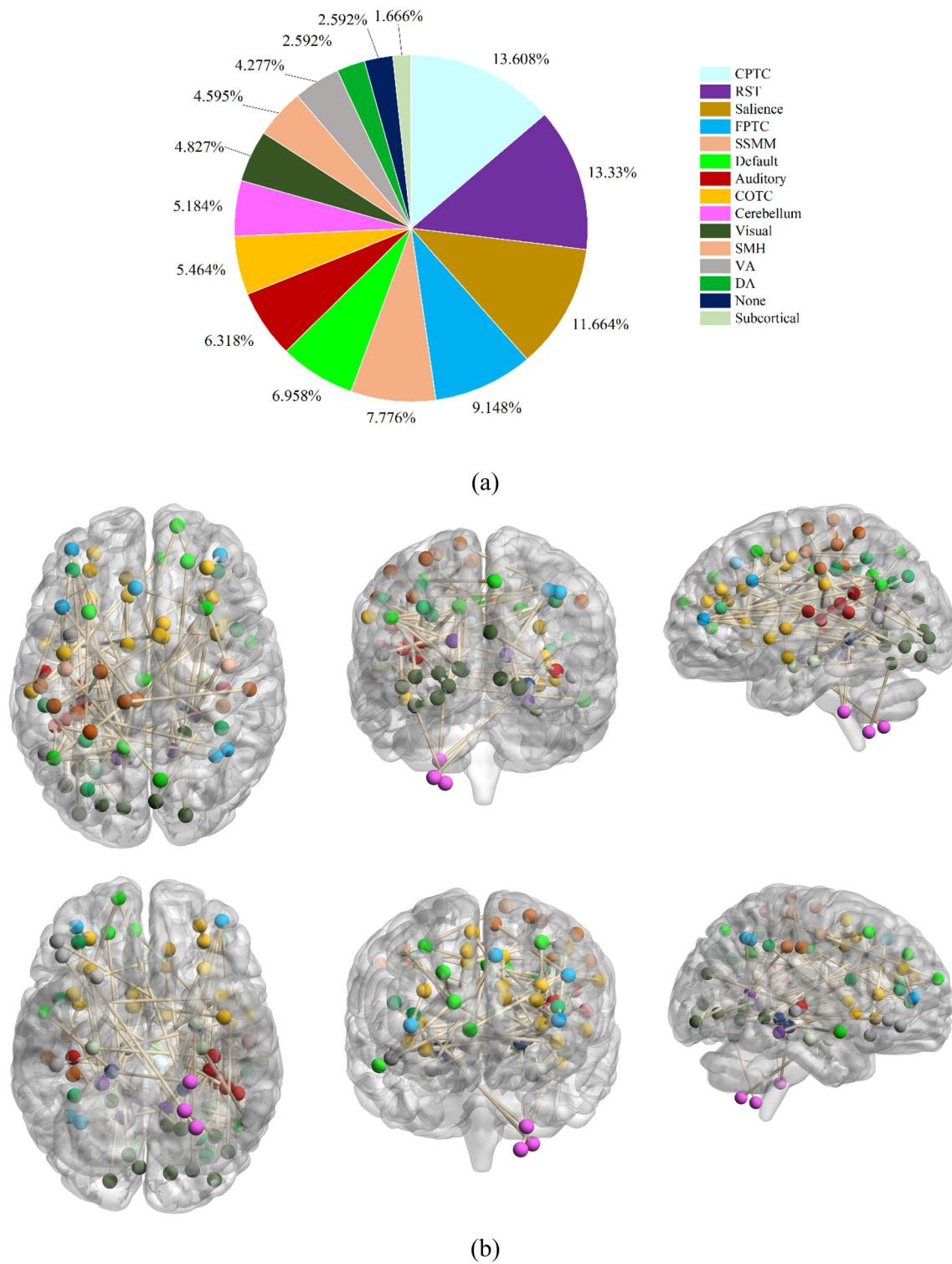
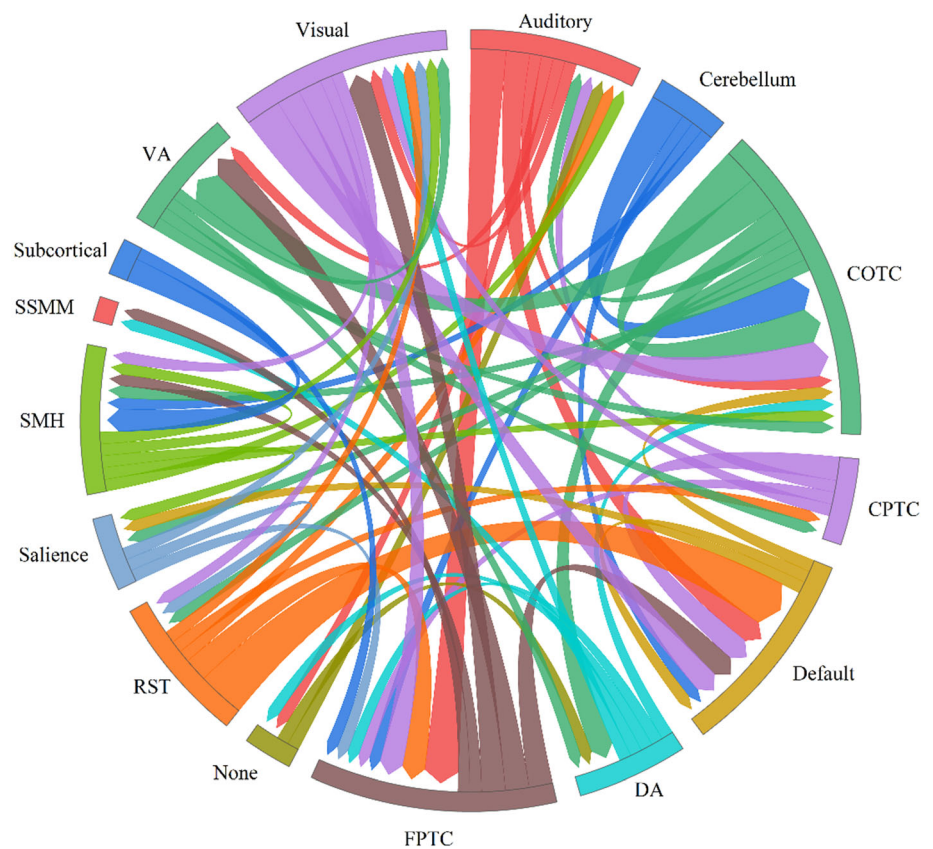


Fig. 7 **a** Pie chart representation of significant 0.3% non-fractal FC features from the brain networks used in the MLP model for the OHSU site
b 3D representation of top features contributed to the OHSU site

Fig. 8 Connectome representation of brain connections that helped to discriminate ASD from TD on the OHSU site



non-fractal FC, and MLP produced the highest fivefold average classification accuracy of 100%. In comparison with PCC and fractal FC, it outperformed other machine learning models. There have been several studies that use site-specific [64] as well as global [8] classification models to diagnose ASD using a combination of FC and machine learning methods. The covariance FC method was compared with machine learning and deep learning algorithms for the classification of ASDs in a few studies [31]. In most studies, PCC was used as part of a machine learning pipeline or deep learning pipeline [86]. In a recent study [28], non-fractal FC and SVM performed better than PCC with SVM at diagnosing autism.

3.8 Limitations and future scope

In our study, we found that non-fractal FC has significant potential for identifying a biomarker for the diagnosis of autism spectrum disorders. The method must be tested on a larger data set and with a global classification model before it can be used in clinical settings. Several factors may affect the non-fractal connectivity method, including the length of the fMRI BOLD series and the wavelet transform. Our current approach is successful in many ways, but there are some limitations we need to address in the future.

- The present work has only considered the machine learning classifiers, and we have not implemented deep learning techniques for clinical diagnosis. Because it requires a great deal of data and computation resources, it can pose a challenge in healthcare settings. In the future, we intend to use lightweight deep-learning models to improve our clinical diagnosis accuracy.
- We should also acknowledge that our methodology has been tested only on a limited sample size, comprising 311 autistic people and 400 typically developed people. Therefore, it needs to be validated on other datasets to ensure its effectiveness and reliability.
- Thirdly, we have not classified all sites as was done in many earlier works. Future work will include investigating the features (fractals and non-fractals) of all sites to diagnose ASD as well as reducing the computational burden on the system.
- Our study relied on MRI technology, which has inherent limitations in spatial and temporal resolution and may affect FC measurements. Moreover, methodological choices in fMRI data preprocessing and analysis could introduce biases, underscoring the need for standardization in future research.
- In terms of symptomatology and underlying neurobiology, ASD is a highly heterogeneous disorder. As a result of this variability, it is challenging to develop

Table 5 Comparison of the performance of present work with a list of earlier works which used the ABIDE dataset for ASD diagnosis

Refs.	Database	Single site/multiple Sites	No. of atlas / ROI	Feature extraction	Feature selection	Classification	Performance measure (max accuracy in %)
[78]	ABIDE I	Single (KKI)	1/215	Fractal	XGBoost	LR	94.7
[79]	ABIDE II	Single (AAL)	1/96	Spectral features	N/A	N/A	N/A
[80]	ABIDE I	Single (BASC)*	1/122	Nine connectivity measures	N/A	SVM, RF, NB, LR, MLP, CNN	99.00 (RF & LR)
[81]	ABIDE I	Single (AAL)	1/116	Progressive Fourier transform and pre-trained CNN model features	N/A	SVM, KNN	96.7 (KNN)
[82]	ABIDE I	Single (AAL)	1/116	Pretrained networks based on CNN	N/A	SVM, KNN	85.9 (KNN)
[67]	ABIDE I	Multiple sites (12 sites)	1/116	PCC	N/A	Few shot learning and Siamese Networks	80.76 (OLIN – Max accuracy)
[83]	ABIDE I (structural and functional MRI data)	Single	N/A	Edges and spatio-temporal features	N/A	3D-CNN	81.23
[28]	ABIDE I	Single (AAL)	1/116	PCC and non-oscillatory connectivity measure	ANOVA <i>p</i> -value-based feature ranking	SVM	98.6 (ASD Vs NC) 88.9 (ASD, APD**, PDD***, NC)
[84]	ABIDE II (structural MRI)	Single (NYU)	N/A	Volume- and surface-related features	Fast correlation-based feature selection	SVM, RF, KNN, ANN	86.29 (SVM) 71.15 (RF) 86.53 (KNN) 88.46 (ANN)
[85]	ABIDE I	Multiple (PITT, YALE, NYU, UM)	1/122	Graph connectivity-related features	N/A	DNN	66.66 (tenfold) 80 (Leave one out)
Ours	ABIDE I and ABIDE II	Both single and multiple	3/236	Fractal and non-fractal	XGBoost	SVM, MLP, NB, SGD, RF, LR, BC, KNN	97.17 (MLP)—All sites 100 (MLP)—Single site

*BASC: bootstrap analysis of stable clusters ** APD: Asperger disorder *** PDD: pervasive development disorder

consistent biomarkers, and there is a need for tailored approaches in research and clinical practice.

The implications for future research and clinical applications are as follows:

- Future studies could improve clinical diagnosis accuracy with the implementation of lightweight deep-learning models. However, this approach is likely to yield more refined insights despite its challenges in data and computational resource requirements.
- A longitudinal study and application of our methodology to a wide variety of datasets are crucial next steps. In this way, we will be able to track ASD progression over time and validate our findings with a variety of demographic profiles and clinical characteristics.
- A detailed analysis of the FC patterns observed offers promising prospects for developing personalized treatment strategies. The correlation between variations in these patterns and symptom severity and therapeutic response should be investigated in future research.
- By studying neurodevelopmental trajectories in autism spectrum disorders and applying these methods to other neurodevelopmental disorders, we can gain a deeper understanding of these conditions.
- Efforts should be directed toward making advanced neuroimaging techniques more accessible and feasible in clinical settings. The integration of these methods

into routine diagnostic procedures would involve overcoming practical challenges.

4 Conclusion

This study investigated the potential of non-fractal FC methods and fractal FC methods for diagnosing autism spectrum disorders using fMRI data and machine learning algorithms. We evaluated the performance of three FC methods, PCC, fractal correlation, and non-fractal correlation, in site-specific classification models for ASD diagnosis compared to most single-site and single-atlas diagnoses. We achieved a fivefold average classification accuracy of 100% using the MLP algorithm at the OHSU site. There were significant differences between ASD and TD based on the CPTC network (13.66%), RST network (13.33%), and Salience network (11.66%). Moreover, non-fractal FC performs better than PCC and fractal methods, eliminating fractal components caused by system noise and physiological processes. The use of non-fractal connectivity allowed more accurate measurement of spontaneous brain activity, improving the ability to distinguish individuals with ASD from those with TD. The use of FC measures based on fractals and non-fractals in the diagnosis of ASDs is a completely new technique in resting-state fMRI. As a result of our study, it is important to select an FC measure that is appropriate for the classification of neurodevelopmental disorders. In this study, we provide valuable insight into the application of fractal and non-fractal FC methods for ASD diagnosis, which contributes to the development of more accurate and reliable diagnostic tools in the fields of neuroimaging and machine learning. This research may lead to early intervention for individuals suffering from autism spectrum disorders and related disorders and improved outcomes in the future.

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Data availability The datasets (ABIDEI and ABIDEII) generated during and/or analyzed during the current study are available as open-source data in https://fcon_1000.projects.nitrc.org/indi/abide/abide-II.html.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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Authors and Affiliations

Chetan Rakshe¹ · Suja Kunneth² · Soumya Sundaram³ · Murugappan Murugappan^{4,5,6}  · Jac Fredo Agastinose Ronickom¹

✉ Murugappan Murugappan
m.murugappan@kcst.edu.kw

¹ Computational Neuroscience and Biology Lab, School of Bio-Medical Engineering, Indian Institute of Technology (BHU), Varanasi, Uttar Pradesh, India

² National Institute of Speech & Hearing, Thiruvananthapuram, Kerala, India

³ Sree Chitra Tirunal Institute for Medical Sciences and Technology, Thiruvananthapuram, Kerala, India

⁴ Intelligent Signal Processing (ISP) Research Lab, Department of Electronics and Communication Engineering, Kuwait College of Science and Technology, Block 4, Doha, Kuwait

⁵ Department of Electronics and Communication Engineering, School of Engineering, Vels Institute of Sciences, Technology, and Advanced Studies, Chennai, Tamil Nadu, India

⁶ Center of Excellence for Unmanned Aerial Systems (CoEUAS), Universiti Malaysia Perlis, 02600 Perlis, Malaysia